

Project Contribution Statement

Project Title: Real-Time Crime Analytics & Intelligent Alert System **Course:** Fundamentals of Big Data Analytics (CS4109)

Group Details

- **Member 1:** Sameed Ahmed Siddiqui (22i-8223)
- **Member 2:** Junaid Zeb (22i-8794)

Contribution Breakdown by Component

Component	Sameed Ahmed Siddiqui	Junaid Zeb
Batch Layer (Apache Spark & MLlib)	60%	40%
Speed Layer (Kafka & Apache Storm)	55%	45%
Serving Layer (PostgreSQL & MongoDB)	40%	60%
Infrastructure & Docker Setup	55%	45%
Dashboard Development (Streamlit)	35%	65%
Technical Documentation & Report	45%	55%

Task Details

- **Sameed Ahmed Siddiqui:** Responsible for the implementation of the Storm Topology bolts (WindowBolt, AnomalyBolt), Kafka producer simulation, Spark batch analytics, and Windows PowerShell orchestrator (run.ps1).
- **Junaid Zeb:** Responsible for dataset preprocessing, schema enforcement in PySpark, SQL/NoSQL schema design, Streamlit dashboard visualization, Docker Compose files and configuration, and the comprehensive LaTeX technical report.

Signatures

Sameed Ahmed Siddiqui

Date: May 8, 2026

Junaid Zeb

Date: May 8, 2026